

Name: _____

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



(Est. 1911)

MATHEMATICS

YEAR 11

PRELIMINARY ASSESSMENT TASK 2

July 2005

Time Allowed – 70 mins

DIRECTIONS TO CANDIDATES:

- Write your solutions in ink.
- Approximate marks are shown in each question. Marks may be adjusted slightly during the marking process if necessary.
- All working should be shown in every question.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Each question attempted is to be started ON A NEW PAGE, clearly marked with the number of the question and your name and class on the top right hand side of the page.

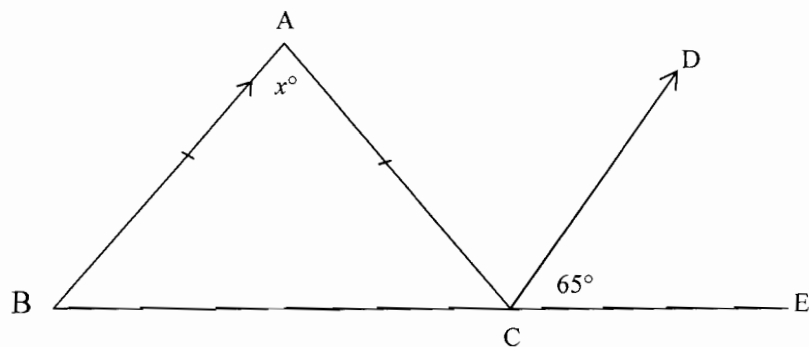
QUESTION	MARK
Question 1	
Question 2	
Question 3	
Question 4	
Question 5	
Question 6	
Question 7	
Question 8	
TOTAL	

Question 1 (7 marks)

- a) Solve $2 \sin A = \sqrt{3}$ for $0^\circ \leq A \leq 360^\circ$ 2
- b) State the domain for $y = \frac{\sqrt{x+1}}{x-1}$ 2
- c) Sketch the region where $y \leq 4 - x^2$ 1
- d) i) Sketch $y = |x + 4|$, $-5 < x < 1$ 1
- ii) Hence or otherwise, write down the solution for $|x + 4| = -x$ 1

Question 2 (7 marks) START A NEW PAGE

- a) Find the exact value of $\sec 60^\circ$ 1
- b) In the diagram $AB = AC$ and $AB \parallel DC$ 3
 Giving a reason for each step, find the value of x .



- c) Find the exact solution of $\tan^2 A - 3 = 0$ if $0^\circ \leq A \leq 360^\circ$ 3

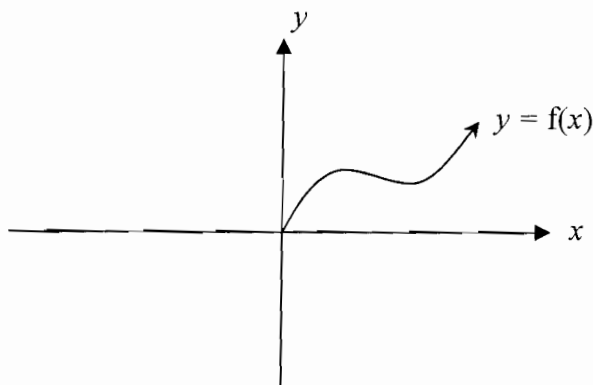
Question 3 (7 marks) **START A NEW PAGE**

The points A (-3, 6), B (-6, 0) and C form a right angled triangle with the right angle at A. The point C is on the y axis.

- a) Draw a neat sketch showing this information. 1
- b) Find the exact length of AB. 1
- c) Find the slope of AB. 1
- d) Find, in general form, the equation of AC (do not attempt to find the y intercept yet) 2
- e) Hence find the x and y intercepts of the line through A and C. 2

Question 4 (7 marks) **START A NEW PAGE**

- a) $y=f(x)$ has been drawn for $x \geq 0$ 2



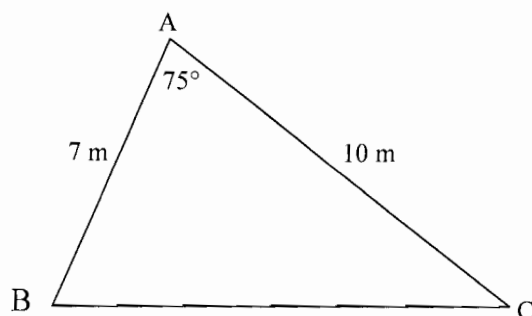
- i) Copy the diagram and complete the graph of $y = f(x)$ if $f(x)$ is an EVEN function
 - ii) Copy the diagram and complete the graph of $y = f(x)$ if $f(x)$ is an ODD function
- b) Sketch the region defined by the inequations $x + y \leq 1$ and $x + y \geq 0$ 3
- c) Prove that $\frac{\cos^2 \theta}{(1 - \cos \theta)(1 + \cos \theta)} = \cot^2 \theta$ 2

Question 5**(7 marks)****START NEW PAGE**

- a) If $\tan A = -\frac{3}{8}$ and $\cos A < 0$ find the exact value of $\sin A$

2

b)



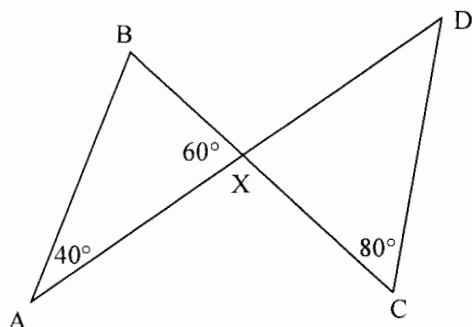
- i) Find the length of BC to the nearest cm.

3

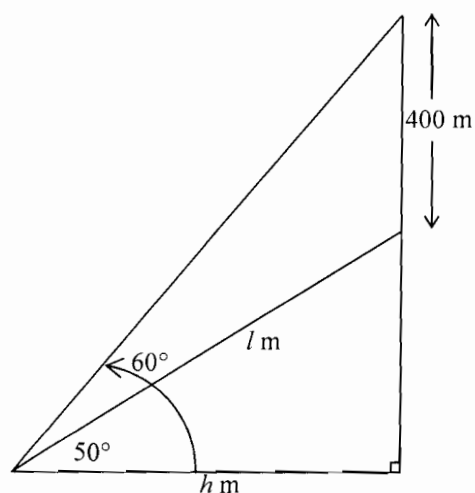
- ii) Find the area of $\triangle ABC$ (in m^2 to 2 dec. places)

2**Question 6****(7 marks)****START A NEW PAGE**

- a) Lines AD and BC intersect at X. Prove that AB is parallel to CD.

2

b)



- i) Use the sine rule to find the value of l correct to 1 decimal place.

3

- ii) Use this value of l to find the value of h correct to 1 dec. place.

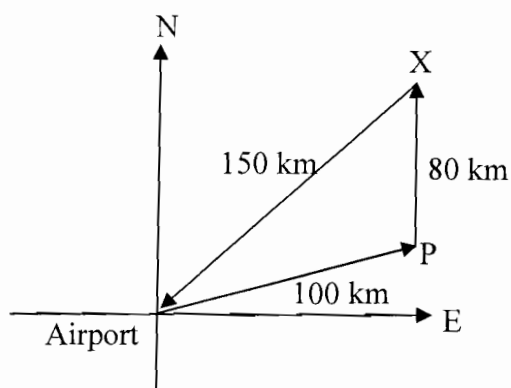
2

Question 7**(7 marks)****START A NEW PAGE**

- a) Determine the sum of the interior angles of a regular polygon if each exterior angle is 24° . **2**
- b) i) From $\sin^2 \theta + \cos^2 \theta = 1$ obtain another Pythagorean Identity for $\operatorname{cosec}^2 \theta$ (show your reasoning clearly). **2**
- ii) Hence simplify $1 - \cot^2 \theta + \operatorname{cosec}^2 \theta$ **1**
- c) Find the equation of the line which is parallel to the line $2x - 5y + 3 = 0$ and contains the point $(1, 4)$. **2**

Question 8**(7 marks)****START A NEW PAGE**

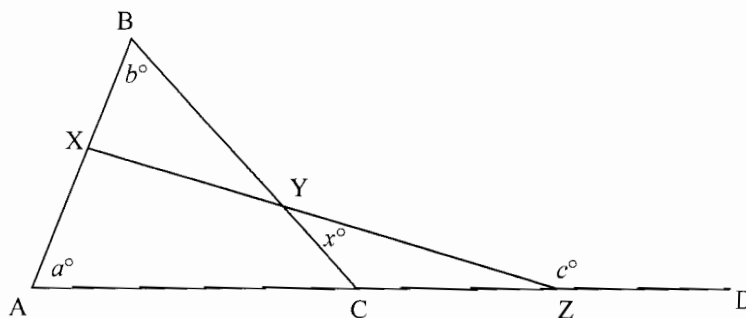
- a) **5**



A pilot flies out from the airport on a bearing which is roughly a bit north of east for 100 km to point P. She then turns due north and flies for 80 km to point X. At this point she is exactly 150 km from the airport.

- i) Calculate the bearing to the nearest minute on which she must fly to return directly to the airport.
- ii) Write down the bearing of point X from the airport.

- b) **2**



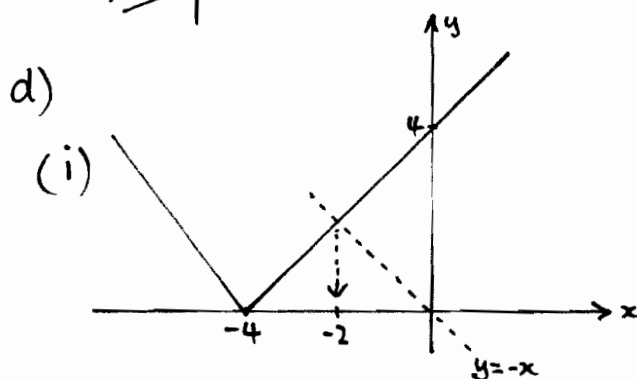
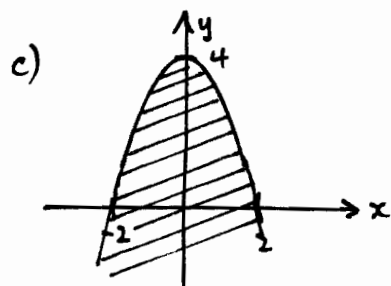
AD, XZ, AB and BC are straight lines. Find an expression for x in terms of a , b and c .

YEAR 11 2 UNIT ASSESSMENT, TASK 2 JULY 2005

QUESTION 1

a) $\sin A = \frac{\sqrt{3}}{2}$
 $\therefore A = 60^\circ$ or 120°
 ① ①

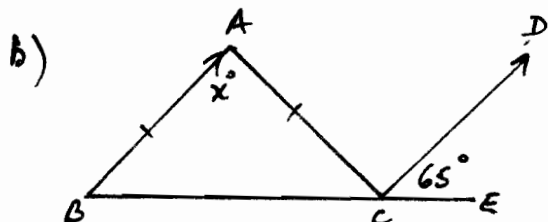
b) $x \neq 1, x \geq -1$



(ii) $x = -2$

QUESTION 2

a) $\sec 60^\circ = \frac{1}{\cos 60^\circ} = 2$



$\angle B = 65^\circ$ (Corresponding $\angle$ s, $AB \parallel DC$)
 $\angle ACB = 65^\circ$ ($\angle$ s opposite equal sides)
 $x + 65 + 65 = 180$ ($\angle$ sum of $\triangle ABC$)
 $\therefore x = 50$

c) $\tan^2 A = 3 \Rightarrow \tan A = \pm\sqrt{3}$
 $\therefore A = 60^\circ$ or 120°

SOLUTIONS AND MARKING SCHEME.

1 for each value

1 mark each

1 or 0. At least x or y intercepts correct.

i) 1 or 0. x intercept must be shown. $y = -x$ need not be shown.

(ii) ① No working required.

a) ① mark

① statement + reason

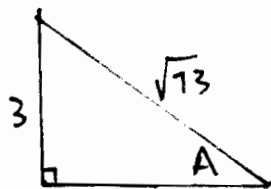
① " "

① " "

All 4 answers correct - 3 marks
 60° and 240° only - 2 marks

QUESTION 5

(a) A is in 2nd quadrant

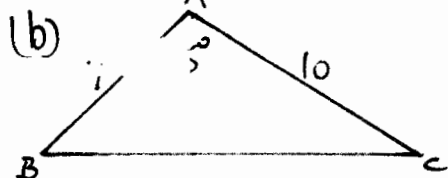


$$\sin A = \frac{3}{\sqrt{13}}$$

Correct answer - 2 marks

or

1 mark for choosing 2nd quadrant



(i) $BC^2 = 7^2 + 10^2 - 2 \cdot 7 \cdot 10 \cdot \cos 75^\circ$

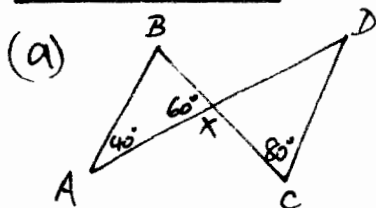
$\therefore BC = 10.62 \text{ m}$

3 marks if correct or
2 marks if incorrect rounding (2 d.p.)

1 mark for correct formula

(ii) $\text{Area} = \frac{1}{2} \cdot 10 \cdot 7 \cdot \sin 75^\circ$
 $= 33.81 \text{ m}^2$

QUESTION 6



$40^\circ + 60^\circ + \angle B = 180^\circ$

$\therefore \angle B = 80^\circ$ (L sum of $\triangle ABX$)

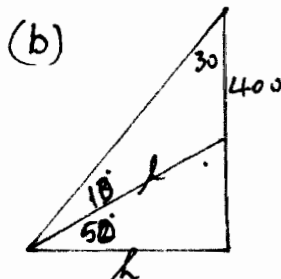
$\therefore \angle B = \angle C$

$\therefore AB \parallel CD$ (alternate Ls equal)

1 mark for showing $\angle B = 80^\circ$
or
 $\angle D = 40^\circ$

AND

1 mark for correct reason
for parallelism



(i) $\frac{l}{\sin 30^\circ} = \frac{400}{\sin 10^\circ}$
 $\therefore l = \frac{400 \cdot \sin 30^\circ}{\sin 10^\circ}$

≈ 1151.75

(i) 3 for correct answer - or
2 for correct formula or
1 for finding 30°

(ii) $\frac{h}{l} = \cos 50^\circ$
 $h = 1151.75 \times \cos 50^\circ$
 ≈ 740.3

(ii) 2 for correct answer (~ 740)
 - allow more exact answer
 if better value of l used
 from (i). OR

QUESTION 7

a) $24n = 360^\circ$

$\therefore n = 15$

$\angle \text{Sum} = 15 \times 156^\circ = 2340^\circ$

b) i) $\sin^2 \theta + \cos^2 \theta = 1$

ie $\frac{\sin^2 \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$

ie $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

ii) $1 - \cot^2 \theta + \operatorname{cosec}^2 \theta$
 $= 1 - \cot^2 \theta + 1 + \cot^2 \theta$ from (i)
 $= 2$

c) Parallel line has eqn

$2x - 5y + c = 0$

$x=1, y=4: 2 - 20 + c = 0$
 $\therefore c = 18$

Reqd Eqn: $2x - 5y + 18 = 0$

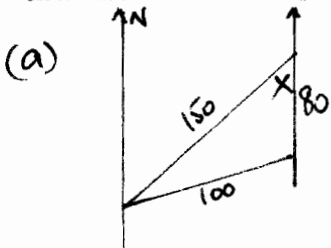
2 marks for correct answer OR
 1 mark for $n=15$

2 marks for correct answer OR
 1 mark if working not shown.

1 mark

2 marks for correct answer OR
 1 for showing $2x - 5y + c = 0$ OR
 1 for showing slope is $\frac{2}{5}$

QUESTION 8



(i) $100^2 = 150^2 + 80^2 - 2 \cdot 150 \cdot 80 \cos X$

ie $\cos X = \frac{150^2 + 80^2 - 100^2}{2 \cdot 150 \cdot 80}$

$= 38^\circ 3'$ ————— (d)

$\therefore$ Course to Airport $218^\circ 3'$

(ii) X bears $38^\circ 3'$ from Airport

(b)

$x = c - a - b$

3 for correct answer OR
 2 for correct formula for $\cos X$ OR
 1 for attempting to find $\angle X$

1 mark if same as (d)

2 for correct answer OR
 1 if an exterior $\angle$ calculated correctly